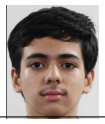
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PROJECT AND TEAM INFORMATION

Project Title

SmartCare: Next-Gen Digital Healthcare

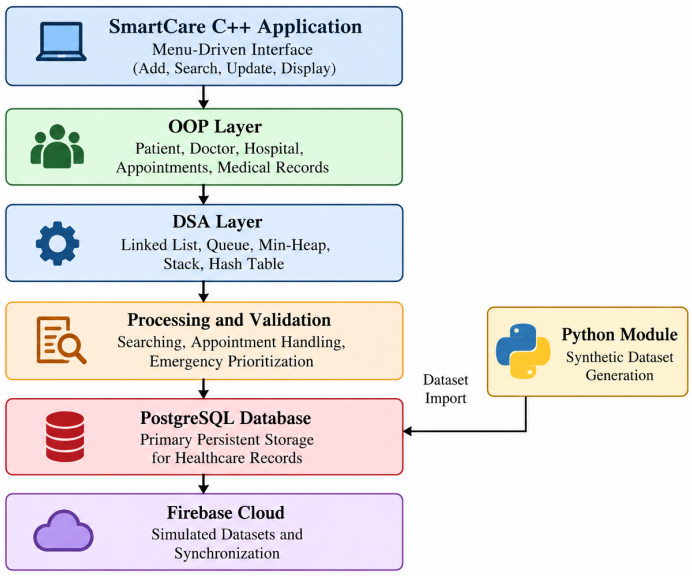
Team Name: *Team #*	*SMART CARE* *DSCPP-III-2026-T256*
Team member 1 (Team Lead)	*Rana, Rishabh – 2028043* *2510012020@geu.ac.in* 
Team member 2	*Panwar, Mantram Singh – 2028760* *2510378345@geu.ac.in* 
Team member 3	*Rawat, Sarthak Singh – 2029011* *2510350026@geu.ac.in* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

In Phase 2, we worked on bringing the main parts of SmartCare together. Our project is a C++ application that helps manage patient details, doctors, appointments, medical records and emergency cases. We used OOP concepts and data structures like Linked List, Queue, Min-Heap, Stack and Hash Table for different tasks.
We connected the application to PostgreSQL and added 200 synthetic patient records. Python is used to prepare sample data for testing. Firebase integration is still pending, along with testing on larger datasets and adding controls for sharing records between hospitals. We will work on these parts in the next phase. SmartCare is currently a college project that uses simulated data, not a system for real medical use.

Updated Project Approach and Architecture (2 pts)



Tasks Completed (7 pts)

Task Completed	Team Member
Created OOP classes for patients, doctors, hospitals, records, and appointments	Rishabh Rana
Linked List for patient records and Queue for appointment scheduling	Rishabh Rana
Min-Heap for prioritising emergency cases	Sarthak Singh Rawat
Stack for keeping recent-action history	Sarthak Singh Rawat
Hash Table for searching patients by SmartCare ID	Mantram Singh Panwar
PostgreSQL table design and connection with C++	Mantram Singh Panwar
Python synthetic-data generator and menu-driven interface	Mantram Singh Panwar

Challenges/Roadblocks (7 pts)

- **Database connection:** We had to check that C++ was connecting properly to PostgreSQL and that patient records could be saved and retrieved.
- **Data structures:** We needed to check that patient searching, appointment handling, emergency priority and action history worked as expected.
- **Dataset preparation:** The 200 sample patient records needed to follow the correct format before being added to PostgreSQL.
- **Firebase integration:** We have not completed the Firebase connection yet. Data mapping and synchronization are still left to do.
- **Combining modules:** Getting all the parts to work together requires testing and fixing errors wherever they appear.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Complete Firebase synchronization	Sarthak Singh Rawat
Add access control for sharing records between hospitals	Mantram Singh Panwar
Testing on larger synthetic datasets	ALL TEAM MEMBERS
Final documentation and presentation	ALL TEAM MEMBERS

Project Outcome/Deliverables (2 pts)

- We have built the main C++ SmartCare application with a menu to access different healthcare features.
- We used OOP and DSA concepts to manage patient records, appointments, emergency cases and other operations.
- We connected PostgreSQL and imported 200 synthetic patient records.
- We created a Python module to prepare sample healthcare data for testing.
- We are using GitHub to store our code and keep track of project updates.
- Firebase integration, hospital record-sharing controls and testing with larger datasets are still remaining.

Progress Overview (2 pts)

- We have completed the main C++ modules and connected the application to PostgreSQL.
- We have imported 200 synthetic patient records and prepared sample data using Python.
- We still need to test all modules together and check how the system works with more records. Firebase integration and hospital record-sharing controls are also pending.
- Our next step is to complete these features, fix any issues found during testing, and finish the report and presentation.

Codebase Information (2 pts)

- **GitHub Repository:**https://github.com/mantram10/DSCPP-III-T256-SMART$_CARE_PROJECT$
- **Version Control:** Git helps us track changes and share updates with the team.
- **C++ Code:** Contains the main application, OOP classes and DSA operations.
- **PostgreSQL:** Contains the database setup files and table-related SQL scripts, where included.
- **Python Module:** Used to generate synthetic patient records for testing.
- **README:** Explains the project and setup steps; we will keep it updated as the code changes.
- **Project Updates:** We push code changes to GitHub as we complete and check our work.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
Patient search by ID (Hash Table)	Implemented; testing pending	Run searches with an existing ID and an invalid ID.
Appointment booking (Queue)	Implemented; testing pending	Check that appointments follow FIFO order and are saved correctly.
Emergency handling (Min-Heap)	Implemented; testing pending	Confirm that cases are processed according to their priority values.
Recent-action history (Stack)	Implemented; testing pending	Check that the newest action is displayed first.
Synthetic data load into PostgreSQL	Pass	200 synthetic patient records imported
Firebase synchronization	Pending	Not connected in the current application
Large-dataset testing	Pending	Further performance testing planned

Deliverables Progress (2 pts)

- Core OOP classes and DSA operations – **Completed**
- PostgreSQL tables and application connection – **Completed**
- Python script for generating synthetic data – **Completed**
- Import of 200 synthetic patient records – **Completed**
- Reading patient records from PostgreSQL – **Completed**
- Firebase connection and synchronization – **Pending**
- Testing the integrated application – **In progress**
- Controls for sharing records between hospitals – **Pending**
- Final report and presentation – **In progress**